**14** $y$ is inversely proportional to $x^3$

$y = 44$ when $x = a$

Show that $y = 5.5$ when $x = 2a$

$$y = \frac{k}{x^3}$$

**1 mark**
Write the inverse-cube equation.

When $x = a$: $44 = \frac{k}{a^3}$

**1 mark**
Substitute the given pair.

so $k = 44a^3$.

When $x = 2a$: $y = \frac{44a^3}{(2a)^3}$

$$y = \frac{44a^3}{8a^3} = 44 \div 8 = 5.5$$

**1 mark**
Show the substitution, cancellation and stated result.

**(Total for Question 14 is 3 marks)**